Information Perception and Environmentally Conscious Behavior

—An Evaluation Using Conjoint Analysis—

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<Table of Contents>

1 Introduction

1.1 Research Background

1.2 Research Objectives

1.3 Research Methodology

2 Empirical Study

2.1 Survey Data

2.2 Changes in Preferences Across the Entire Sample

2.3 Effects of Environmental Information via Print Media

2.4 Effects of Environmental Information via Other Media

3 Discussion

3.1 Effects of Differences in Media Information Content

3.2 Effects of Environmental Information by Individual Media Type

4 Conclusion

Endnotes

References

1 Introduction

1 . 1 Research Background

When considering the reduction of environmental impact and the maintenance of physical health through environmentally conscious consumer behavior, information asymmetry problems are always present, stemming from the fact that the environmental and health premiums inherent in environmentally friendly products cannot be discerned simply by visual inspection or touch at the time of purchase. Therefore, it is important to properly present environmental and quality information about products through product labels and other means, thereby correcting information asymmetry and enabling appropriate consumption choices. In this regard, it cannot necessarily be said that higher objectivity and reliability of the provided environmental and quality information leads to higher consumer trust in that information; the “reliability” of the information itself and the “trust” that consumers place in it should be considered separately[1)]. Trust, in particular, has the characteristic of being extremely difficult to form while being easily broken. In today’s context, where trust in certification information has been severely undermined by a series of fraudulent labeling incidents, there is an urgent need to restore this trust. Moreover, previous psychological and behavioral economics research has also indicated that trust in information sources is important for information regarding food safety and the environment (Frewer et al. 1996, Hunt and Frewer 2001), making it a critical task to examine methods of information

provision that can further enhance consumer trust.

On the other hand, regarding the manner of information provision, there are two positions: one advocating that as much detailed information as possible should be provided, and another arguing that information should be provided after appropriate selection and integration. Recentempirical studies using stated preference methods have demonstrated that excessive amounts of information hinder accurate choice-making[2)]2)Furthermore, considering actual purchasing behavior, it is unrealistic to assume that consumers can always properly process complex and vast amounts of information and make correct choices.

Therefore, it is necessary to consider more effective methods of information provision within an appropriate range of information volume. To this end, it is essential to identify which factors contribute to the formation of consumer trust during information provision. Peters et al. (1997) have summarized the conditions for trust formation as including factors such as the knowledge and expertise of the information source, the openness and honesty of the information source, and concern and care. In this regard, psychological and economic research has demonstrated that the strength of trust varies depending on the information source[3)]3), and several studies have shown that trust in third-party certification by consumer organizations is the highest, while trust in certification by businesses and government is relatively low (Huffman et al. 2004, Sato et al. 2005).

In this study, in order to analyze the effects of factors other than

2)

3)

certification organizations as information sources, print media-based information on food safety and environmental issues is provided periodically, and the frequency of exposure to environmental information through other media is surveyed. The study attempts to identify the factors necessary for more effective information provision to promote environmentally conscious behavior.

1．2 Research Objectives

In this study, we first attempt to empirically verify the following two hypotheses. The firsthypothesis posits that regularly providing information on food safety and environmental issues increases preferences for environmentally friendly products compared to cases where no information is provided. The second hypothesis posits that higher frequency of obtaining information on food safety and environmental issues from media such as newspapers and television increases preferences for environmentally friendly products compared to cases with lower frequency.

Furthermore, this study aims to examine the analytical results for these two hypotheses, distinguish between factors that are and are not effective in promoting trust in environmental and quality information, and provide discussion on effective methods of information provision. The ultimate intention is that the information asymmetry problem in the market for environmentally friendly products will be corrected, consumer trust in environmental and quality information will be restored, and thereby environmentally conscious behavior will be promoted.

1．3　Research Methodology

In this study, as an example of environmentally friendly products, we focus on vegetables, which have the highest purchase frequency among daily necessities[3)]. The data used is consumer behavior data surveyed in 2006[4)]. This includes data primarily identifying which factors among product attributes such as price and freshness consumers focus on when purchasing vegetables (one bunch of spinach), as well as detailed surveys of consumer personal attributes and awareness of environmental issues.

First, using a portion of this data, consumers' willingness to pay (WTP) for spinach origin, quality assurance labels, and freshness is calculated using a method called choice-based conjoint analysis. Next, after providing information on environmental impact and food safety a total of four times, willingness to pay is calculated again using the same procedure. Using the data obtained through this process, changes in consumer preferences before and after information provision are examined. Furthermore, by dividing subjects into groups based on consumer attributes and whether or not information was provided, changes in consumer preferences for each group are examined and compared, allowing discussion of the various factors that influence changes in preferences.

Choice-Based Conjoint Analysis[5)]

Choice-based conjoint is a method for evaluating product attributes by having

respondents select the most desirable product from among multiple product profiles, developed by Louviere and Woodworth (1986). Compared to CVM (Contingent Valuation Method) and rating-based conjoint, it has the advantage that its response format is closer to actual consumption behavior and is easier for respondents[1]。

In this paper, within the framework of choice-based conjoint analysis, estimation is performed using two methods: the Conditional Logit Model (hereinafter abbreviated as CL model) and the Nested Logit Model (hereinafter abbreviated as NL model). The adjusted R^2 (adjusted R-square) and the Akaike InformationCriterion(Akaike's Information Criterion = AIC) are used as indicators to compare which model more appropriately explains the variation in the data, and the model with the higher goodness of fit is adopted. The CL model is based on the following concept.

product profile j, the total utility when selecting U_j is assumed as the followingrandom utility model.

$$U_j = V_j(x_j, p_j) + \varepsilon_j \qquad (1)$$

where $V_j(x_j, p_j)$ is the observable component of utility, ε_j is the unobservable component, x_j is the attribute vector of product j, p_j is the price of product j.

At this time, the probability that respondent selects product $C(1,2,\cdots,m)$ from the set of products j P_j is the probability that the utility of selecting product j U_j exceeds the utility of selecting any other product k U_k, and is given by the following equation.

1

$$P_j = \Pr(U_j > U_k, \forall k \in C)$$
$$¿ \Pr(V_j - V_k > \varepsilon_k - \varepsilon_j, \forall k \in C)$$
$$¿ \int_{-\infty}^{Vj-V1} \int_{-\infty}^{Vj-V2} \cdots \int_{-\infty}^{Vj-Vm} f(\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \cdots, \tilde{\varepsilon}_m, \Omega) d\tilde{\varepsilon}_m d\tilde{\varepsilon}_{m-1} \cdots d\tilde{\varepsilon}_1 \quad (2)$$

where $\tilde{\varepsilon}_k = \varepsilon_k - \varepsilon_j$ is the difference in error terms, Ω is the covariance matrix, f (.) is the joint density function.

McFadden (1974) showed that when the error terms in (2) are assumed to follow a Type I extreme value distribution (Gumbel distribution), the probability P_j is

$$P_j = \frac{e^{\lambda V_j}}{\sum_k e^{\lambda V_j}} \quad (3)$$

obtained by this equation. Here, λ is the scale parameter, which is typically normalized to 1. Parameter estimation is performed by the maximum likelihood method using equation (3). This model by McFadden is called the multinomial logit or conditional logit model, but it has the limitation of being unable to express flexible substitution patterns among alternatives. The model that relaxes this constraint is the nested logit model. “Nested” refers to hierarchical decision-making, meaning mutually exclusive subsets of the choice set[6)]6). In this model, compared to alternatives in different nests, alternatives within the same nest have error terms that are more similar to each other. Therefore, while the IID condition is imposed within nests as in the conditional logit, the IID condition is relaxed between nests, enabling the expression of more flexible substitution patterns. In the nested logit model, estimation is performed by adding a variable called the inclusive value (IV) parameter for each nest.

6)

Estimation of Willingness to Pay

First, consider the following main effects model for the utility function.

$$V(x,p)=\sum_i \beta_i x_i+\beta_p P \tag{4}$$

where β are the parameters estimated by the conditional logit or nested logit model. Taking the total differential of (4),

$$\sum_i \frac{\partial V}{\partial x_i} dx_i+\frac{\partial V}{\partial p} dp=dV \tag{5}$$

is obtained. Here, fixing the utility level at its initial level $(dV=0)$, and fixing all product attributes x_1 other than product attribute $(dx_i=0, i\neq 1)$ at their initial levels. Under this assumption, the marginal willingness to pay when product attribute x_1 increases by one unit is

$$MWTP_{x_1}=\frac{dp}{dx_1}=-\frac{\partial V}{\partial x_i}/\frac{\partial V}{\partial p}=-\frac{\beta_1}{\beta_p} \tag{6}$$

obtained by this equation.

By estimating the willingness to pay under each condition using the above procedure and treating this as the degree of preference for each attribute, we attempt to verify the two hypotheses.

2 Empirical Study

2．1 Survey Methodology

The survey data used in this study was collected as follows. During the approximately one-month survey period from April 9 to May 19, 2006, Co-op Kobe (Consumers' Cooperative) distributed questionnaire forms to shoppers at both the Ashiya and Nishinomiya stores. The 305 subjects who agreed to participate in the experiment were asked to fill in mailedquestionnaire forms once a week for a total of five times and return the forms by the specified deadline. These subjects were divided into two groups of 150 and 155 persons, and only the latter group received print media-based information provision once a week for a total of four times.

第１表　提供された情報の内容（『今週の解説』）

第一回	「残留農薬と健康リスク」
第二回	「農業と環境問題」
第三回	「食品表示問題と信頼確保」
第四回	「ごみ問題とリサイクル」

A total of five weekly questionnaires were administered during the survey period, asking detailed questions about actual vegetable purchasing behavior, including subjects' exposure to media information and the content of such information.

Subjects were presented with a conjoint experiment in which they were asked to choose one option from a total of four choices: three hypothetical types of spinach combining the four attributes shown in Table 2 at random levels, plus a "would not buy any" option.

第２表　ほうれん草の属性

産地	地場産、国内産、中国産
ラベル	環境NPO認証ラベル、JAS有機認証ラベル、フードプラン・ラベル、ラベルなし
鮮度	新鮮、普通、少し萎びている
価格	100円、130円、160円、190円

This experiment was repeated five times immediately before the survey period and another five times immediately after the survey period, thereby attempting to analyze the effects of information provision and media exposure during the survey period.

2.2 Changes in Preferences Across the Entire Sample

First, the estimation results by the CL model and NL model for the entire sample before and after the survey period are shown in Table 3.

Table 3: Estimation Results for the Entire Sample by CL and NL Models

CLモデル

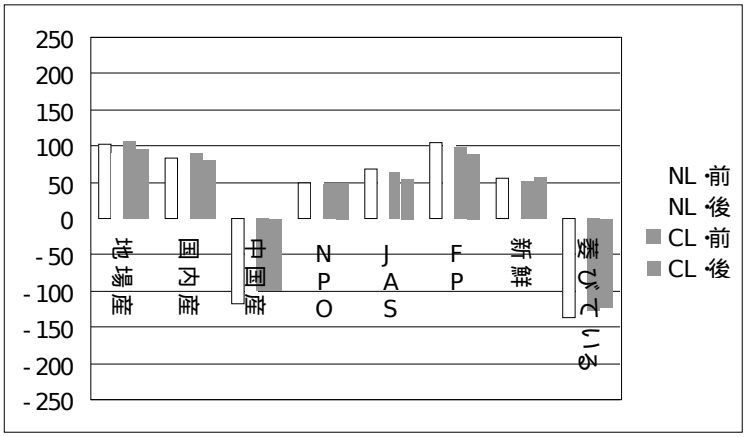
	調査前		調査後	
	係数	t値	係数	t値
地場産	2.579976	9.308	2.591175	8.754
国内産	2.194608	8.049	2.191922	7.483
中国産	- 2.43198	- 6.941	- 2.64628	- 6.922
価格	- 0.02429	- 10.68	- 0.02697	- 10.568
NPO	1.118826	8.423	1.302049	9.253
JAS	1.576814	7.656	1.505638	6.558
FP	2.370983	10.574	2.443193	9.903
新鮮	1.226246	9.51	1.552124	10.841
萎びている	- 3.12829	- 14.342	- 3.27768	- 12.457
選択しない	- 0.83827	- 3.067	- 0.71659	- 2.452
サンプル数	1525, skipped 228 ba		1525, skipped 309 bad obs.	
対数尤度	- 1066.48		- 963.257	
修正済R2	0.47015		0.47502	

NLモデル

	調査前		調査後	
	係数	t値	係数	t値
地場産	2.18081	7.49332	1.75279	7.21506
国内産	1.78291	6.07432	1.28699	5.05389
中国産	- 2.50692	- 7.60691	- 2.61405	- 7.91195
価格	- 0.02134	- 8.77807	- 0.01912	- 8.02052
NPO	1.05935	8.478	1.08469	9.34858
JAS	1.47052	7.53904	1.36169	6.77878
FP	2.22885	10.1606	2.19587	9.63782
新鮮	1.17031	9.52279	1.41251	11.1549
しなびてい	- 2.91917	- 12.7403	- 2.6846	- 10.2626
選択しない	- 0.7814	- 2.45414	- 0.2717	- 0.60945
IV(BUY)	1.28548	9.32184	1.93776	9.01995
IV(NO)	Fixed	1.00E+10	Fixed	1.00E+10
サンプル数	1525, skipped 228 ba		1525, skipped 309 bad obs.	
対数尤度	- 1063.82		- 944.259	
修正済R2	0.47147		0.48538	

Furthermore, calculating the respective willingness to pay values yields the results shown in Figure 2.

Figure 2: Changes Across the Entire Sample by CL and NL Models



Examining Figure 2, slight differences between the CL model and NL model can be observed depending on the attribute. The decision of which model to adopt is based on the comparison of goodness of fit shown in Table 4.

第4表　適合度の比較

修正済みR2

	調査前	調査後
CLモデル	0.47015	0.47502
NLモデル	**0.47147**	**0.48538**

AIC

	調査前	調査後
CLモデル	2152.964	1946.513
NLモデル	**2149.634**	**1910.518**

From Table 4, the adjusted R^2 values are higher and the AIC values are lower for the NL model in both cases, so the NL model is adopted here. However, the differences in these values are all marginal. In Figure 2, regarding the evaluation of quality labels, the CL model shows a slight increase while the NL model shows a slight decrease, but considering that the goodness of fit of both models is nearly identical, even if the NL model is adopted, it is difficult to conclusively state that “preferences for labels decreased slightly,” and it

should rather be considered that “no significant change was observed.” Based on the above reasoning, while preferences for local and domestic produce and freshness appear to have increased slightly, it can be judged that no significant changes were observed for other attributes.

Do these results mean that information provision had virtually no effect? Next, by comparing the preferences of the group that received information with those of the group that did not, we analyze the effects of information provision in greater detail.

2.3 Effects of Environmental Information via Print Media

Table 5 shows the NL model estimation results before and after the survey period for both the group that received information provision and the group that did not.

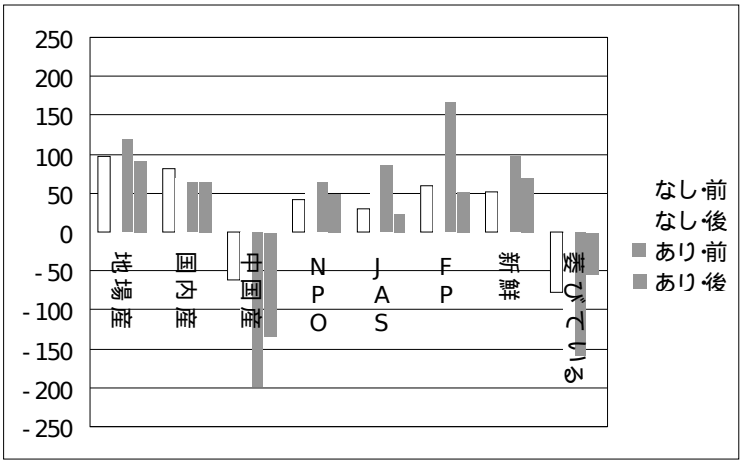
Table 5: Estimation Results by Presence or Absence of Information Provision

	情報提供なし 調査前			情報提供なし 調査後			情報提供あり 調査前			情報提供あり 調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	2.551504	0.636719	4.007	1.242124	0.777673	1.597	2.124125	0.848945	2.502	2.060461	0.542911	3.795
国内産	2.119421	0.689869	3.072	1.347949	1.733203	0.778	1.109467	1.199088	0.925	1.479883	0.577492	2.563
中国産	-1.61182	0.73115	-2.205	-2.28334	2.466037	-0.926	-3.51082	1.850274	-1.897	-2.99957	1.060597	-2.828
価格	-0.02631	0.006571	-4.003	-0.01967	0.005691	-3.456	-0.01771	0.004256	-4.16	-0.02255	0.005982	-3.769
NPO	1.093792	0.245213	4.461	1.306156	0.33208	3.933	1.125585	0.363145	3.1	1.127822	0.183341	6.152
JAS	0.804727	0.703361	1.144	1.224657	0.342189	3.579	1.526433	0.407451	3.746	0.519093	0.56948	0.912
FP	1.545773	0.746393	2.071	1.666044	1.340832	1.243	2.948378	1.004193	2.936	1.162162	0.562814	2.065
新鮮	1.37757	0.216652	6.358	1.078437	1.185604	0.91	1.719571	0.880758	1.952	1.547438	0.22952	6.742
萎びている	-2.03647	0.764593	-2.663	-3.71778	0.623734	-5.961	-2.82031	0.377529	-7.47	-1.2231	0.615391	-1.988
選択しない	-1.19187	0.511239	-2.331	-0.81671	1.234888	-0.661	-0.61853	0.766764	-0.807	-0.6987	0.868533	-0.804
IV(BUY)	1.302121	0.370924	3.51	1.593233	1.593233	1.297	0.429591	0.429591	2.265	2.068512	0.472053	4.382
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	750, skipped 141 bad obs.			750, skipped 136 bad obs.			775, skipped 87 bad obs.			775, skipped 173 bad obs.		
対数尤度	-547.528			-423.505			-513.339			-512.681		
修正済R2	0.38496			0.56347			0.54038			0.40327		

Furthermore, the respective willingness to pay values are calculated and

shown in Figure 3.

Figure 3: Changes by Presence or Absence of Information Provision



From Figure 3, preferences for locally produced goods decreased slightly regardless of whether information was provided or not. On the other hand, for quality label preferences, the group that did not receive information showed slight increases across all labels, while the group that did receive information showed marked decreases across all labels. Also regarding freshness preferences, the group without information provision appears to have become more sensitive. This is a result that nearly reverses the first hypothesis: “Regularly providing information on food safety and environmental issues increases preferences for environmentally friendly products compared to cases where no information is provided.” Why did the group that received information lower its preferences for environmental labels and freshness?

The content of all four issues of the provided information (“This Week’s Commentary”) covered a wide range of topics as shown in Table 1, with no focused theme. The texts prioritized the provision of accurate knowledge over

conveying a clear message, and at the same time, no concrete solutions to current problems were presented. Thesepoints were also frequently noted in questionnaire responses from subjects, and these may have dispersed readers' preferences. Additionally, the third commentary pointed out that the current quality labeling system is not sufficiently reliable. Considering that this lowered preferences for quality labels in general, the estimation results of the effects of information provision can be regarded as natural.

However, the cause of preference changes may not be limited to the print media information provision in this study. Consumers were also exposed to external media such as newspapers, television, and the internet during the survey period, and the effects of these media also need to be analyzed. Therefore, we next examine the second hypothesis: "When the frequency of obtaining information on food safety and environmental issues from media such as newspapers and television is high, preferences for environmentally friendly products increase compared to cases with low frequency."

2.4 Effects of Environmental Information via Other General Media

First, based on data regarding media exposure obtained from the questionnaire, the total number of times subjects obtained information on food safety or environmental issues from any media including newspaper/magazine articles, television, the internet, or word-of-mouth during the survey period was calculated. Those with a total of 0 or more but fewer than 5 times were classified as the low-exposure-frequency group, and those with 5 or more times as the high-exposure-frequency group, dividing the entire sample into two

groups according to the frequency of exposure to environmental information from media. The NL model estimation results before and after the survey period for each of these two groups are shown in Table 6.

Table 6: Frequency of Exposure to Environmental Information via Media and Estimation Results

	低頻度 調査前			低頻度 調査後			高頻度 調査前			高頻度 調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	2.123396	0.383751	5.533	1.841338	0.361243	5.097	2.129758	0.434851	4.898	1.75659	0.342502	5.129
国内産	1.935854	0.384163	5.039	1.323841	0.371782	3.561	1.507433	0.444206	3.394	1.31787	0.361066	3.65
中国産	- 2.31036	0.462525	- 4.995	- 2.61013	0.46825	- 5.574	- 2.75966	0.474098	- 5.821	- 2.72884	0.493739	- 5.527
価格	- 0.02462	0.003497	- 7.04	- 0.02239	0.003835	- 5.84	- 0.01729	0.003393	- 5.098	- 0.01711	0.003105	- 5.511
NPO	0.799414	0.168383	4.748	0.944926	0.166102	5.689	1.366579	0.191765	7.126	1.213458	0.165598	7.328
J AS	1.269927	0.282074	4.502	1.364063	0.316525	4.309	1.721544	0.274979	6.261	1.3908	0.263891	5.27
FP	1.960511	0.318822	6.149	2.405033	0.367294	6.548	2.561797	0.310702	8.245	2.092597	0.294085	7.116
新鮮	1.030246	0.172685	5.966	1.432597	0.197091	7.269	1.337629	0.176289	7.588	1.423281	0.168454	8.449
萎びている	- 2.76581	0.323837	- 8.541	- 2.76648	0.376123	- 7.355	- 3.03551	0.325662	- 9.321	- 2.65096	0.367126	- 7.221
選択しない	- 1.59948	0.459625	- 3.48	- 1.40943	0.65333	- 2.157	0.073023	0.475436	0.154	0.595941	0.603566	0.987
IV(BUY)	1.368462	0.209081	6.545	1.911697	0.336915	5.674	1.2515001	0.18671	6.703	1.895921	0.272133	6.967
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	785, skipped 182 bad obs.			785, skipped 251 bad obs.			740, skipped 46 bad obs.			740, skipped 58 bad obs.		
対数尤度	- 512.264			- 419.122			- 541.425			- 518.126		
修正済R2	0.44511			0.48639			0.50049			0.48865		

Furthermore, the calculated willingness to pay values are shown in Figure 4.

Figure 4: Changes by Frequency of Exposure to Environmental Information via Media

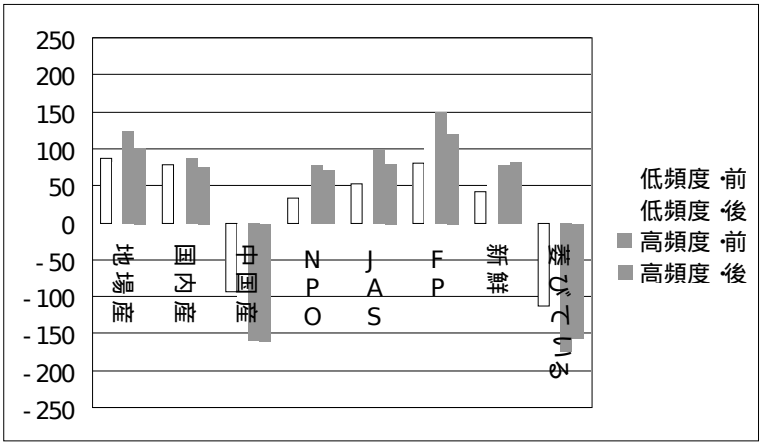


Figure 4 shows the changes in preferences for both the group with low frequency and the group with high frequency of exposure to information on food safety or environmental issues from any media including newspaper/magazine articles, television, the internet, or word-of-mouth. This reveals that, regardless of whether media exposure frequency is high or low, preferences for locally and domestically produced goods tend to decrease, while preferences for freshness have increased slightly. Furthermore, regarding preferences for quality labels, the low media exposure frequency group shows a slight increase, whereas the high-frequency group conversely shows a tendency to decrease. Thus, it can be said that the second hypothesis also does not hold.

3 Discussion

Why did media exposure to environmental information have almost no effect on preferences for origin and freshness, and even worked in the direction of slightly lowering preferences for quality labels? To investigate the cause of this more deeply, we classify the content of media-based environmental information into two categories—information related to food safety and information related to other general environmental issues—and compare the changes in preferences.

3 . 1 3.1 Effects of Differences in Media Information Content

Table 7: Frequency of Exposure to Information on Food Safety and Estimation Results

	低頻度 調査前			低頻度 調査後			高頻度 調査前			高頻度 調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	2.143503	0.367037	5.84	1.992054	0.346683	5.746	2.274282	0.498362	4.564	1.56658	0.358471	4.37
国内産	1.917823	0.374096	5.127	1.523643	0.355959	4.28	1.584373	0.491834	3.221	1.080952	0.380683	2.84
中国産	- 2.4382	0.436647	- 5.584	- 2.68631	0.444908	- 6.038	- 2.62408	0.519654	- 5.05	- 2.71824	0.534307	- 5.087
価格	- 0.02475	0.003435	- 7.205	- 0.02368	0.003662	- 6.466	- 0.01649	0.003547	- 4.649	- 0.01518	0.003273	- 4.637
NPO	0.997024	0.163131	6.112	0.974283	0.154396	6.31	1.188043	0.209443	5.672	1.22511	0.182322	6.72
J AS	1.491681	0.272911	5.466	1.459482	0.292787	4.985	1.477893	0.295139	5.007	1.312286	0.288393	4.55
FP	2.109826	0.305789	6.9	2.395142	0.339462	7.056	2.509379	0.334657	7.498	2.058667	0.317654	6.481
新鮮	1.151105	0.164482	6.998	1.418309	0.184435	7.69	1.276863	0.194264	6.573	1.456357	0.182469	7.981
萎びている	- 2.75515	0.311446	- 8.846	- 2.80135	0.348354	- 8.042	- 3.11612	0.346944	- 8.982	- 2.61195	0.408029	- 6.401
選択しない	- 1.27955	0.428262	- 2.988	- 1.3976	0.605538	- 2.308	- 0.09883	0.482077	- 0.205	0.834471	0.652474	1.279
IV(BUY)	1.362693	0.19578	6.96	1.900658	0.300874	6.317	1.152204	0.189936	6.066	1.901218	0.298683	6.365
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	905, skipped 192 bad obs.			905, skipped 261 bad obs.			620, skipped 36 bad obs.			620, skipped 48 bad obs.		
対数尤度	- 598.606			- 498.831			- 453.568			- 439.812		
修正R2	0.44669			0.49109			0.50976			0.48204		

Table 8: Frequency of Exposure to Information on General Environmental Issues and Estimation Results

	低頻度 調査前			低頻度 調査後			高頻度 調査前			高頻度 調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	1.888068	0.368836	5.119	1.73714	0.350119	4.962	2.495716	0.471457	5.294	1.802636	0.344917	5.226
国内産	1.691895	0.370793	4.563	1.286039	0.365822	3.515	1.86573	0.477927	3.904	1.314018	0.362314	3.627
中国産	- 2.35889	0.444192	- 5.311	- 2.83097	0.488154	- 5.799	- 2.7388	0.496284	- 5.519	- 2.44579	0.453534	- 5.393
価格	- 0.01992	0.003079	- 6.469	- 0.02037	0.003693	- 5.516	- 0.02301	0.004026	- 5.717	- 0.01831	0.003168	- 5.778
NPO	0.782493	0.160403	4.878	0.994469	0.161876	6.143	1.402272	0.202148	6.937	1.177955	0.167243	7.043
J AS	1.200708	0.250295	4.797	1.44325	0.305773	4.72	1.843838	0.314571	5.861	1.324436	0.270982	4.888
FP	1.817109	0.280769	6.472	2.40155	0.351987	6.823	2.77091	0.35631	7.777	2.061066	0.303655	6.788
新鮮	1.043045	0.159227	6.551	1.456424	0.19044	7.648	1.354836	0.196926	6.88	1.397099	0.172299	8.109
萎びている	- 2.6305	0.302544	- 8.695	- 2.71524	0.370921	- 7.32	- 3.24246	0.352834	- 9.19	- 2.67742	0.37008	- 7.235
選択しない	- 0.95292	0.448953	- 2.123	- 0.71341	0.68511	- 1.041	- 0.57938	0.465064	- 1.246	0.100085	0.592144	0.169
IV(BUY)	1.412437	0.215765	6.546	2.006557	0.347248	5.778	1.170167	0.177887	6.578	1.868027	0.276814	6.748
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	830, skipped 187 bad obs.			830, skipped 261 bad obs.			695, skipped 41 bad obs.			695, skipped 48 bad obs.		
対数尤度	- 555.013			- 442.121			- 501.322			- 500.091		
修正済R2	0.44024			0.48598			0.50637			0.484		

Table 7 shows the estimation results before and after the survey period for groups classified into two based on the number of times they were exposed only to food safety-related information among media information. Table 8 shows the estimation results before and after the survey period for groups classified into two based on the number of times they were exposed to

environmental information other than food safety-related information. The willingness to pay values calculated from Tables 7 and 8 are shown in Figures 5 and 6, respectively.

Figure 5: Changes by Frequency of Exposure to Information on Food Safety

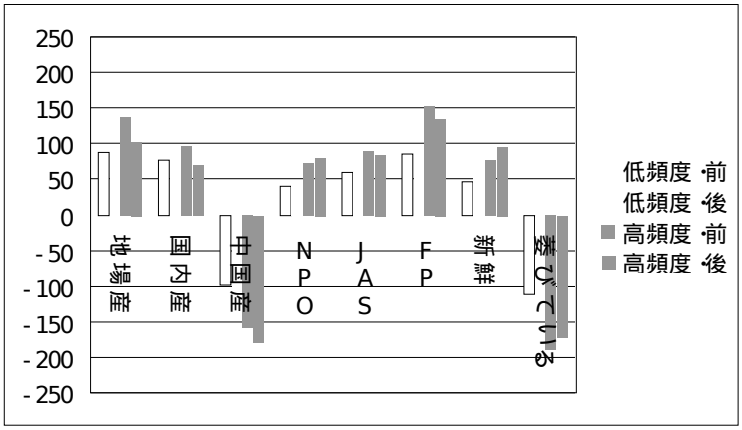
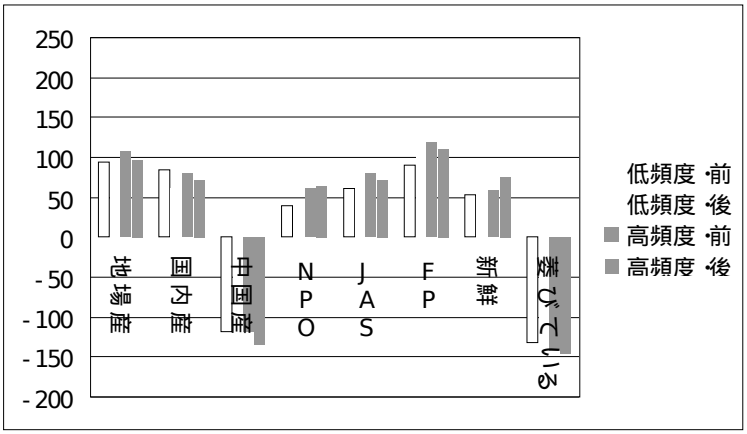


Figure 6: Frequency of Exposure to Information on Environmental Impact



Comparing Figures 5 and 6, it can be seen that there are no significant differences in changes attributable to differences in information content. This

means that the third hypothesis—“When the content of media information relates to food safety, preferences for environmentally friendly products increase more than in the case of other environmental information”—also does not hold.

From the above analytical results, the group with high frequency of media exposure to environmental information, compared to the low-frequency group, was found to have originally higher preferences for locally produced, domestically producedand fresh spinach, as well as quality labels in general, even before the survey period. Conversely, preferences for Chinese-produced or wilted spinach were found to be originally lower before the survey. This can be attributed to the fact that those who frequently obtain information on food safety and environmental issues from media tend to have higher interest and involvement in food and the environment, and tend to value environmentally friendly products more highly. However, the effect of environmental information from media in general on changes in preferences for environmentally friendly products during the survey period was barely observable. Furthermore, when changes in preferences were investigated by the content of media-based environmental information, no significant differences were found either.

3.2 Effects of Environmental Information by Individual Media Type

Therefore, in the following analysis, general media is further classified into three types: “newspaper/magazine articles,” “television,” and “word-of-mouth,”

and the effects of each media type are analyzed individually. First, subjects are divided into two groups by frequency of exposure to environmental information through newspaper/magazine articles. Next, the estimation results before and after the survey period for each group are shown in Table 9. The willingness to pay for each product attribute, calculated based on the coefficient data shown in Table 9, is presented in Figure 7. This procedure is repeated for environmental information from television and word-of-mouth, with the respective estimation results shown in Tables 10 and 11. Furthermore, willingness to pay values calculated from the data in Tables 10 and 11 are shown in Figures 8 and 9, respectively.

Table 9: Frequency of Exposure to Environmental Information via Newspaper/Magazine Articles and Estimation Results

	低頻度 調査前			低頻度 調査後			高頻度 調査前			高頻度 調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	2.420397	0.545182	4.44	2.082094	0.508628	4.094	2.024393	0.351387	5.761	1.730841	0.280375	6.173
国内産	2.43811	0.548228	4.447	1.713002	0.518115	3.306	1.407051	0.358549	3.924	1.227338	0.296118	4.145
中国産	- 1.7888	0.539594	- 3.315	- 2.24351	0.567713	- 3.952	- 3.00477	0.43516	- 6.905	- 2.84239	0.429361	- 6.62
価格	- 0.02696	0.004728	- 5.702	- 0.02399	0.005067	- 4.735	- 0.01845	0.002937	- 6.281	- 0.01881	0.002809	- 6.697
NPO	0.81947	0.221594	3.698	0.881867	0.219386	4.02	1.23309	0.157687	7.82	1.147224	0.139506	8.223
J AS	1.609274	0.403792	3.985	0.988103	0.357584	2.763	1.565248	0.235916	6.635	1.516743	0.253871	5.974
FP	2.071292	0.450763	4.595	1.59567	0.400531	3.984	2.513089	0.269266	9.333	2.453102	0.288047	8.516
新鮮	1.095834	0.22241	4.927	1.275818	0.227053	5.619	1.242027	0.150828	8.235	1.48633	0.156591	9.492
萎びている	- 2.85294	0.406166	- 7.024	- 3.75609	0.663714	- 5.659	- 2.98495	0.282343	- 10.572	- 2.41966	0.292793	- 8.264
選択しない	- 1.46569	0.471577	- 3.108	- 1.50381	0.668371	- 2.25	- 0.27164	0.434055	- 0.626	0.088105	0.577133	0.153
IV(BUY)	1.04588	0.218139	4.795	1.389027	0.266387	5.214	1.395592	0.180007	7.753	2.186847	0.306105	7.144
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	565, skipped 164 bad obs.			565, skipped 205 bad obs.			960, skipped 64 bad obs.			960, skipped 104 bad obs.		
対数尤度	- 352.427			- 286.3			- 698.513			- 650.611		
修正済R2	0.43301			0.47982			0.49584			0.49126		

Figure 7: Frequency of Exposure to Environmental Information via Newspaper/Magazine Articles

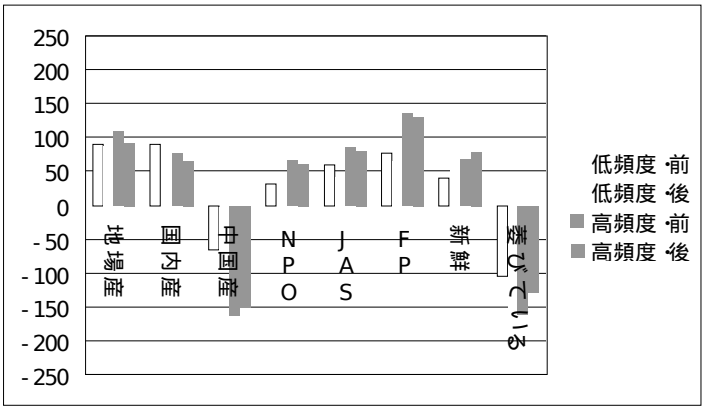


Table 10: Frequency of Exposure to Environmental Information via Television and Estimation Results

	低頻度 ･調査前			低頻度 ･調査後			高頻度 ･調査前			高頻度 ･調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	1.924448	0.471343	4.083	1.873418	0.47468	3.947	2.358356	0.376486	6.264	1.682797	0.287068	5.862
国内産	1.60668	0.502596	3.197	1.556607	0.476901	3.264	1.876634	0.37304	5.031	1.157255	0.306918	3.771
中国産	- 1.71041	0.545609	- 3.135	- 2.24009	0.670898	- 3.339	- 2.80913	0.421681	- 6.662	- 2.77038	0.384631	- 7.203
価格	- 0.02096	0.0046	- 4.556	- 0.02366	0.004764	- 4.966	- 0.02172	.003009	- 7.217	- 0.01761	0.002789	- 6.314
NPO	0.866962	0.207075	4.187	0.957833	0.231655	4.135	1.100218	0.154211	7.134	1.126411	0.134196	8.394
J AS	1.284123	0.36512	3.517	0.801018	0.416854	1.922	1.56399	0.23966	6.526	1.554125	0.231466	6.714
FP	1.55061	0.407639	3.804	1.88262	0.465699	4.043	2.521544	0.26913	9.369	2.310985	0.263534	8.769
新鮮	1.265066	0.231971	5.454	1.334567	0.257718	5.178	1.192694	0.150208	7.94	1.448585	0.145897	9.929
菱びている	- 2.95889	0.565462	- 5.233	- 2.74475	0.548362	- 5.005	- 2.94971	0.256688	- 11.491	- 2.6824	0.29843	- 8.988
選択しない	- 0.93055	0.841717	- 1.106	- 1.49831	0.793325	- 1.889	- 0.73848	0.34277	- 2.154	0.171904	0.559526	0.307
IV(BUY)	2.043417	0.492187	4.152	1.746897	0.330783	5.281	1.102174(	0.139893	7.879	2.041586	0.289529	7.051
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	480, skipped 160 bad obs.			480, skipped 181 bad obs.			1045, skipped 68 bad obs.			1045, skipped 128 bad obs.		
対数尤度	- 261.84			- 227.615			- 791.514			- 713.212		
修正済R2	0.4544			0.49168			0.48175			0.48375		

Figure 8: Frequency of Exposure to Environmental Information via Television

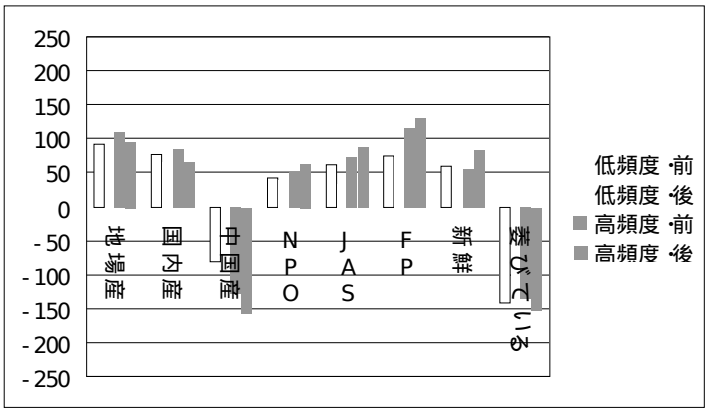
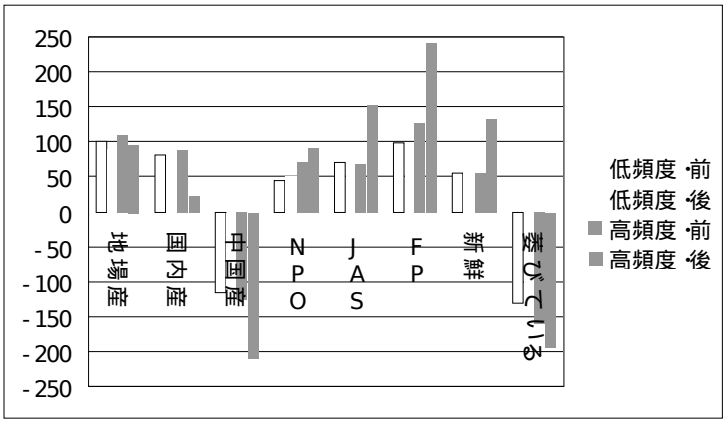


Table 11: Frequency of Exposure to Environmental Information via Word-of-

Mouth and Estimation Results

	低頻度 調査前			低頻度 調査後			高頻度 調査前			高頻度 調査後		
	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値	係数	標準誤差	t値
地場産	2.211149	0.333443	6.631	1.872833	0.304767	6.145	2.069696	0.595442	3.476	1.272752	0.42934	2.964
国内産	1.813988	0.334891	5.417	1.559995	0.315996	4.937	1.657821	0.607235	2.73	0.313058	0.496534	0.63
中国産	- 2.55596	0.384868	- 6.641	- 2.66262	0.40473	- 6.579	- 2.39318	0.641815	- 3.729	- 2.76562	0.607941	- 4.549
価格	- 0.02211	0.002868	- 7.708	- 0.02106	0.002953	- 7.134	- 0.01892	0.004634	- 4.083	- 0.0132	0.00452	- 2.921
NPO	0.966652	0.142247	6.796	1.080091	0.141281	7.645	1.333639	0.260342	5.123	1.209078	0.1971	6.134
J AS	1.533155	0.228575	6.707	1.27916	0.2335	5.478	1.307038	0.375714	3.479	2.005353	0.4179	4.799
FP	2.173707	0.256954	8.46	2.019466	0.262491	7.693	2.400165	0.423659	5.665	3.189987	0.518196	6.156
新鮮	1.216262	0.145336	8.369	1.328481	0.14814	8.968	1.054208	0.231632	4.551	1.759542	0.2398	7.338
萎びている	- 2.87737	0.269761	- 10.666	- 2.90317	0.320279	- 9.065	- 3.04043	0.43856	- 6.933	- 2.55338	0.527358	- 4.842
選択しない	- 0.83622	0.37824	- 2.211	- 0.71778	0.464695	- 1.545	- 0.62297	0.603416	- 1.032	3.069687	2.303237	1.333
IV(BUY)	1.329481	0.167374	7.943	1.661699	0.204571	8.123	1.2057	0.248817	4.846	4.10125	1.278559	3.208
IV(NO)	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10	Fixed	0	1.00E+10
サンプル数	1145, skipped 203 bad obs.			1145, skipped 269 bad obs.			380, skipped 25 bad obs.			380, skipped 40 bad obs.		
対数尤度	- 779.706			- 687.601			- 280.383			- 247.358		
修正済R2	0.46281			0.48172			0.49532			0.50789		

Figure 9: Frequency of Exposure to Environmental Information via Word-of-Mouth



Comparing Figures 7, 8, and 9, no significant effects on preferences are observed for environmental information via newspaper/magazine articles. However, for information from television and especially word-of-mouth environmental information, it can be inferred that there is an effect of increasing preferences for quality labels. The impact of word-of-mouth environmental information is particularly large, with preference increases

sufficient to offset the preference decreases caused by print media information provision. This is considered to sufficiently explain why, despite print media information provision significantly lowering preferences for quality labels, almost no change in preferences was observed for the sample as a whole. Further analysis is needed to investigate more precise causes, but at the very least, among the three representative media types analyzed in this study, it can be said that word-of-mouth is the most effective at improving preferences for environmentally friendly products, particularly quality labels.

Indeed, much research on the effects of word-of-mouth has been conducted in the marketing field and elsewhere, and it has been empirically demonstrated that word-of-mouth exerts a stronger influence than advertising, particularly in cases of high uncertainty such as new products and in cases where prior expectations have not been formed. At the same time, the influence of word-of-mouth is said to correlate with trust formation in information sources through accumulated communication and involvement.[7)]7)

From these findings as well, it is considered that consumer involvement as information recipients and communication between consumers and information providers are important factors for effective information provision.

4 Conclusion

It has been empirically demonstrated that neither of the two initially established hypotheses holds true. The first hypothesis yielded nearly opposite

7)

results, but this can also be considered to provide significant implications for effective information provision. Namely, the fact that preferences for quality labels decreased as a result of reliable information provision highlighting current problems can be interpreted as an expression of intent to review the current quality label system and build a more reliable information provision system. Considering such a possibility, a decline in preferences for current quality labels does not necessarily bring about negative consequences for promoting environmentally conscious behavior, and at the same time, an increase in preferences for current quality labels does not necessarily bring about positive consequences.

Regarding the second hypothesis, although no significant effects from media in general were observed, when examined by individual media type, it was found that word-of-mouth has the greatest influence on improving preferences for quality labels, with some influence also coming from television. This appears to be a highly effective means when considering only the perspective of simply increasing consumer preferences for quality labels. However, as stated earlier, increasing preferences for current quality labels cannot be unconditionally considered to lead to the promotion of environmentally conscious behavior.

What is important is not to make consumers blindly trust, but rather that highly reliable information needed by consumers is trusted in proportion to its reliability. If channels such as word-of-mouth and television have the effect of improving consumer preferences even for unreliable information sources, the magnitude of their influence could be a double-edged sword as a means of information provision to promote environmentally conscious behavior. This is

because the environmental and health premiums of environmentally friendly products are credence attributes[9)]9), which cannot be accurately evaluated solely from communication or experiential history. Part of the evaluation of the quality of environmentally friendly products must depend on environmental and quality information, which is why the reliability of the provided information is important.

Therefore, rather than thinking solely from the perspective of restoring consumer trust or increasing preferences, it is necessary to simultaneously enhance the reliability of the information itself and, bearing in mind that the influence of word-of-mouth correlates with trust formation in information sources through communication and involvement, strive for information provision that encourages proactive consumer participation.

Endnotes

1) Yamagishi (1998) distinguishes between the words “trust” and “trustworthiness,” and Sato (2006) also adopts this concept.

2) Sato, Masayuki, “Measurement and Control of Information Overload Using Stated Preference Methods,” 21COE Interfaces for Advanced Economic Analysis Kyoto University, 2005, pp.4-5.

3) Frewer et al. (1996) and Peters et al. (1997) demonstrated this in a manner

9)

that corroborates psychological theory.

4) Based on the results of the Ministry of Agriculture, Forestry and Fisheries' FY2003 Food Consumption Monitor 3rd Regular Survey.

5) This data is from a survey conducted by the Japan Science and Technology Agency project "Regeneration of Organic Matter Circulation Systems through Urban-Rural Cooperation" (Principal Investigator: Professor Kazuhiro Ueta).

6) Based on the explanation in Kuriyama, Koichi and Ishii, Hiroshi, "Environmental Value and Market Competitiveness of Recycled Products: An Evaluation Using Conjoint Analysis" (1999).

7) Based on the explanation in Yoda, Takanori, "Broadband Economics."

8) From Hamaoka, Yutaka and Katahira, Hidetaka, "Mechanisms of Word-of-Mouth Influence: An Information System Model of the Trust Formation Process" (1997).

9) In contrast to attributes that can be judged by visual inspection or touch before purchase (search attributes) and attributes that can be judged through actual purchase and consumption (experience attributes), credence attributes are characteristics that cannot be judged through suchactions.

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